浙江大学 **2004-2005** 学年冬季学期

《**Physics**（Ⅱ）》课程 期末考试 试卷

开课学院：理学院　　考试形式：闭卷，允许带非存储计算器、纸质词典入场

考试时间 :2005 年 1 月 21 日　　所需时间 120 分钟

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题序	一	二	三	四	五	六	七	八	总分
得分									
评卷人									

Planck constant $h=6.63\times10^{-34}$ J s　　Charge of an electron $e=1.6\times10^{-19}$ C

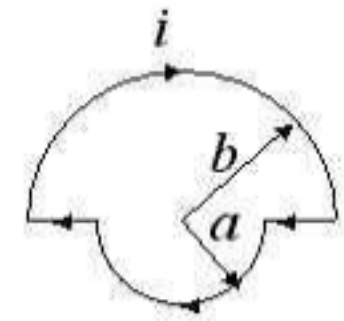
Permittivity constant $\varepsilon_0=8.85\times10^{-12}$ C^2/(N m^2)　Electronic volt 1eV=1.6×10^{-19} J

Permeability constant $\mu_0=4\pi\times10^{-7}$ H/m　　Mass of an electron $m_e=9.11\times10^{-31}$ kg

Ⅰ. Multiple choices (there is one correct answer only):

1. The electric charge on a conductor is
   A. Uniformly distributed throughout the volume.
   B. Confined to the surface and is uniformly distributed.
   C. Mostly on the outer surface, but is not uniformly distributed.
   D. Entirely on the surface and is distributed according to the shape of the object.
   E. Distributed throughout the volume of the object and distributed according to the object's shape.

2.* You are given a closed circuit with radii a and b, as shown in the Fig. , carrying current i. The magnetic dipole moment of the circuit is:



A. $i\pi\frac{(a^2+b^2)}{4}$.

B. $i\pi(a^2+b^2)$

C. $i\pi\frac{(a^2+b^2)}{2}$.

D. $i\pi(a+b)$.

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3. A parallel plate capacitor is attached to a battery that supplies a constant potential difference. While the battery is still attached, the parallel plates are separateda little more. Which statement describes what happens?

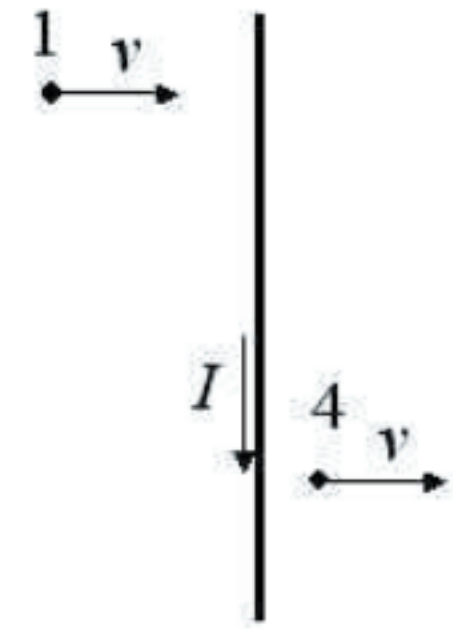
. A. The electric field increases and the charge on the plates increases.

B. The electric field remains constant and the charge on the plates increases.

C. The electric field remains constant and the charge on the plates decreases.

D.Both the electric field and the charge on the plate decreases.

4. A wire carries current as shown in the Fig. Point charge 1 and 4 move in the directions indicated. If the magnetic forces on charges 1 and 4 are equal and their velocities are equal,



A. The charges have the same sign and $|q_1| > |q_4|$.

B. The charges have opposite sign and $|q_1| > |q_4|$.

C. The charges have the same sign and $|q_1| < |q_4|$.

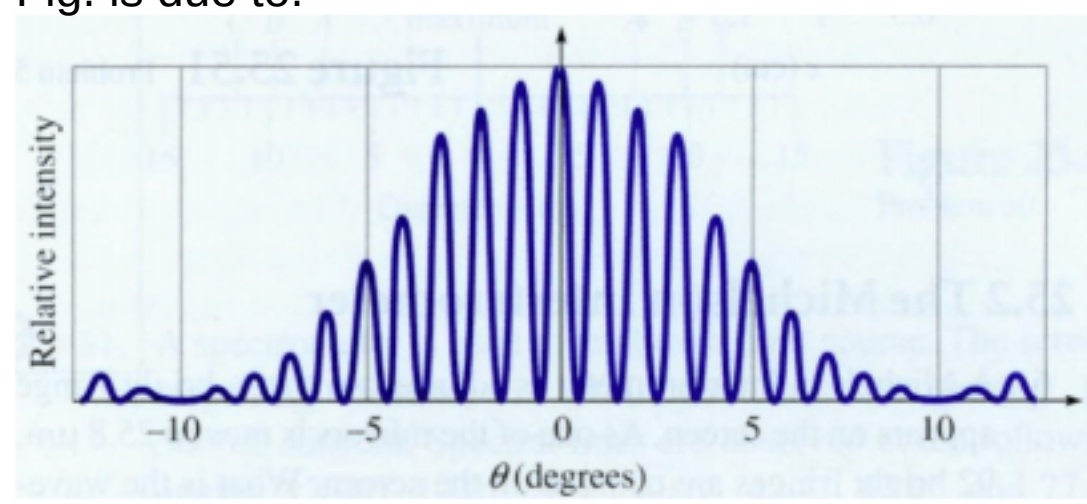
D. The charges have opposite sign and $|q_1| < |q_4|$.

5. Displacement current exists in the region between the plates of a parallel plate capacitor if:

A. the capacitor leaks(泄漏) charge across the plates
B. the capacitor is being discharged
C. the capacitor is fully charged
D. the capacitor is fully discharged
E. none of the above are true

6. The intensity pattern as shown in Fig. is due to.

A. two slits
B. a single slit
C. a grating
D. a circular aperture
E. none of above



7. The Fermi energy of copper at
0K represents

A .the average energy of the electrons

B. the energy necessary to remove an electron from the surface.

C.the maximum energy of any electron.

D. the minimum energy of any electron.

8. How many of Oxygen eight electrons are found in the 3p state?

A. 0
B. 2
C. 4
D. 6
E. 8

9. In the Compton effect, a photon of wavelength λ and frequency f is scattered from an electron, initially at rest. In this process,

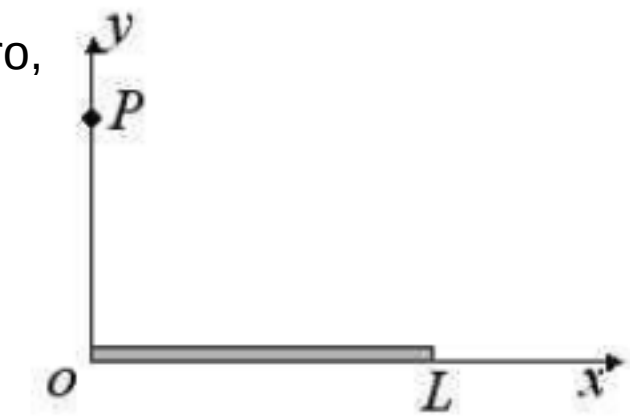
A. the electron gains energy from the photon so that the scattered photon's wavelength is less than λ.
B. the electron gives energy to the scattered photon so that the photon's frequency is greater than f.
C. momentum is not conserved, but energy is conserved.
D. the photon loses energy so that the scattered photon has a frequency less than f.

10. Specify which of the following types of materials might satisfy this description: a filled valence band, an empty conduction band, an energy gap of 0.5eV.

A. Conductor
B. Insulator
C. Semiconductor
D. None of above

II .Calculation problems (present the necessary equations in solution):

1.* On a thin rod of length L lying along the x axis with one end at the origin (x=0), as in Figure, there is distributed a charge per unit length given by $\lambda = kr$, where k is a constant and r is the distance from the origin.



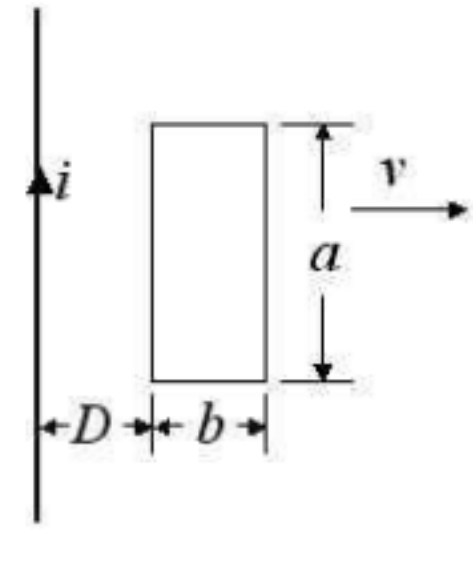
(a) Taking the electrostatic potential at infinite to be zero, find V at the point P on the y axis.

(b) Determine the vertical component, E_y, of the electric field at P from the result of part (a).

(c) Why cannot E_x, the horizontal component of the electric field at P, be found using the result of part (a)?

(d) At what distance from the rod along the y axis is the potential equal to one-half the value at the left end of the rod?

2.*A rectangular loop of wire with length a , width b, and resistance R is placed near an infinite long wire carrying current i, as shown in Figure. The distance from the long wire to the loop is D. Find



(a) the magnitude of the magnetic flux through the loop.

(b) the current in the loop as it moves away from the long wire with speed v.

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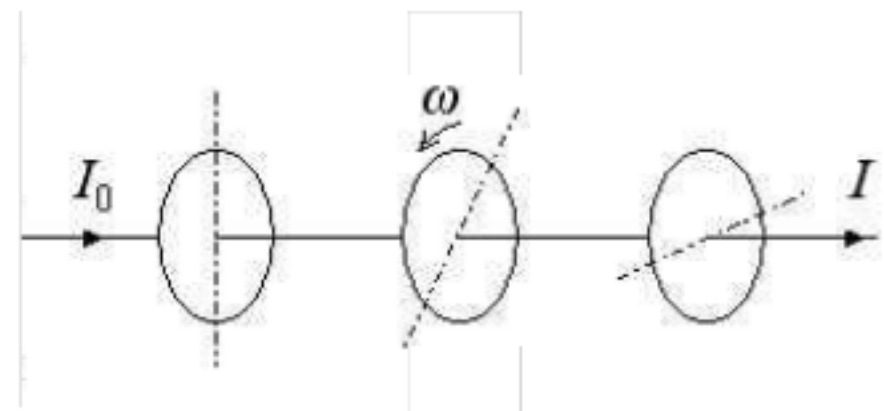
3. A film of soapy water in air is held vertically and viewed in reflected light. The film has index of refractionn=1.36.

(a) Explain why the film appears black at the top.

(b) The light reflected perpendicular to the film at a certain point is missing the wavelengths 504mm and 630nm. No wavelengths between these two are missing. What is the thickness of the film at that point?

(c) What other visible wavelengths are missing, if any?

4. Show in the figure. Suppose that the transmission axes of the left and right polarizing sheets are perpendicular to each other. Also, let the center sheet be rotated on the common axes with an angular speed ω. Show that if unpolarized light is incident on the left sheet with an intensity I_0, the intensity of the beam emerging from the right sheet is



$$I = \frac{1}{16} I_0 (1 - \cos 4\omega t)$$

5. A laser printer puts tiny dots of ink (toner) on the page. The dots should be sufficientl y close together (and therefore small enough) that we don ’t see individual dots, rather, we see letters or graphics. Approximately how many dots per inch(dpi) ensure that you don ’t see individual dots when viewing a page 0.40m from the eye in bright light? Use a pupil(瞳孔) diameter of 2.5mm and λ≒400nm, and 1inch=25.4mm.

6.* Calculate the wavelength of a 100-keV electron and of a bullet of mass 41g travels at 960m/s. Why does the wave nature of the bullet not reveal itself through diffraction effect?

7. For $l=2$, find

(a)the magnitude of the angular momentum $\mathbf{L}$.

(b)the possible values of m_l.

(c)Draw to scale a vector diagram showing the possible orientations of $\mathbf{L}$ with the z axis.

8. The wavefunction of 2**s** state of the hydrogen atom is

$$\psi_{nlm_l} = \psi_{200} = \frac{1}{4\sqrt{2\pi}}\left(\frac{1}{a_0}\right)^{\frac{3}{2}}\left(2-\frac{r}{a_0}\right)e^{-r/2a_0} \qquad \text{where } a_0 = \frac{\varepsilon_0 h^2}{\pi m e^2}$$

(a) Find the principle quantum number, orbital quantum number and magnetic quantum number respectively.

(b) Find the value of the probability density at $r=a_0$.

(c) Find the value of the radial probability density at $r=a_0$.

Give your answer in terms of a_0.